

# 5. Data Warehousing and Online Analytical Processing

Knowledge Discovery in Databases with Exercises

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- 1. Data Warehouse: Basic Concepts**
- 2. Data Warehouse Modeling: Data Cube and OLAP**
- 3. Data Warehouse Design and Usage**
- 4. Data Warehouse and Data Mining**
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# Data Warehouse: Basic Concepts

William “Bill” H. Inmon is commonly referred to as the “father of the data warehouse”.

## Data Warehouse

A data warehouse is a **subject-oriented, integrated, time-variant, and nonvolatile** collection of data in support of management’s decision-making process.<sup>1</sup> Common abbreviations: DW or DWH.

### Other definitions exist:

- “A data warehouse is a system that extracts, cleans, conforms, and delivers source data into a dimensional data store and then supports and implements querying and analysis for the purpose of decision making.”<sup>2</sup>
- A **decision-support** database that is **maintained separately** from the organization’s operational database.
- Supports information processing by providing a solid platform of **consolidated, historical data** for analysis.

**Data warehousing:** The process of constructing and using data warehouses.

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<sup>1</sup>W. H. Inmon, *Building the Data Warehouse*. Wiley, 2005, 4th edition, ISBN: 978-076459-944-6

<sup>2</sup>R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*. Wiley, 2004, ISBN: 978-0764567575

- **Organized around major subjects.** Such as customer, product, sales.
- **Focusing on the modeling and analysis of data for decision makers.** Not on daily operations or transaction processing.
- **Provide a simple and concise view around particular subject issues.** By excluding data that are not useful in the decision-support process.

- **Constructed by integrating multiple heterogeneous data sources.**
  - Relational databases, flat files, online transaction records, . . .
- **Data-cleaning and data-integration techniques are applied.**
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources.
  - E.g., hotel price: currency, tax, breakfast covered.
  - When data is moved to the data warehouse, it is converted.
  - ETL – Extract, Transform, Load.

- The **time horizon** for a data warehouse is **significantly longer** than that of operational systems.
  - Operational database: current-value data.
  - Data warehouse: provide information from a historical perspective, e.g. past 5 — 10 years.
- **Every key structure in the data warehouse contains an element of time, explicitly or implicitly.**
- The key of operational data may or may not contain a "time element."

- **A physically separate store of data.**
  - Transformed from the operational environment.
  - By copying.
- **No operational update of data:**
  - Hence, does not require transaction processing, i.e. no logging, recovery, concurrency control, etc.
  - Requires only three operations:
    1. Initial loading of data.
    2. Refresh (update, often periodically, e.g. over night).
    3. Access of data.



## Three kinds of data warehouse applications.

### 1. Information processing.

- Supports querying, basic statistical analysis, and reporting using crosstabs, tables, charts and graphs.

### 2. Analytical processing.

- Multidimensional analysis of data warehouse data.
- Supports basic OLAP operations such as slicing, dicing, drilling, and pivoting.

### 3. Data mining.

- Knowledge discovery from hidden patterns.
- Supports associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools.

|                           | OLTP                                                     | OLAP                                                               |
|---------------------------|----------------------------------------------------------|--------------------------------------------------------------------|
| <b>Users</b>              | clerk, IT professional                                   | knowledge worker                                                   |
| <b>Function</b>           | day-to-day operations                                    | decision support                                                   |
| <b>DB Design</b>          | application-oriented                                     | decision support                                                   |
| <b>Data</b>               | current, up-to-date; detailed, flat relational; isolated | historical; summarized, multidimensional, integrated, consolidated |
| <b>Usage</b>              | repetitive                                               | ad-hoc                                                             |
| <b>Access</b>             | read/write; index/hash on primary key                    | lots of scans                                                      |
| <b>Unit of Work</b>       | short, simple transaction                                | complex query                                                      |
| <b>#-Records Accessed</b> | 10                                                       | $10^6$                                                             |
| <b>#-Users</b>            | 1000                                                     | 100                                                                |
| <b>DB Size</b>            | 100 MB to GB                                             | 100 GB to TB                                                       |
| <b>Quantification</b>     | transaction throughput                                   | query throughput, response                                         |

OLTP = Online Transaction Processing, OLAP = Online Analytical Processing

## High performance for both systems:

- **DBMS**: tuned for OLTP; Access methods, indexing concurrency control, recovery.
- **Data Warehouse**: tuned for OLAP; Complex OLAP queries, multidimensional view, consolidation.

## Different functions and different data:

- *Missing data (DBMS)*: Decision support (DS) requires **historical data** which operational DBs do not typically maintain.
- *Data consolidation (warehouse)*: DS requires **consolidation** (aggregation, summarization) of data from heterogeneous sources.
- *Data quality (warehouse)*: Different sources typically use inconsistent data representations, codes and formats which have to be reconciled.

## Note

There are more and more systems which perform OLAP analysis directly on relational databases.

## 1. Enterprise Warehouse:

- Collects all of the information about subjects spanning the entire organization.

## 2. Data Mart:

- A **subset** of corporate-wide data that is of value to a **specific group of users**.
- Typically contains (highly) summarized data.
- Independent vs. dependent (directly from warehouse) data mart.

## 3. Virtual Warehouse:

- Also known as *data virtualization*.
- A set of **views** over operational databases.

As an *operational database* are all data sources considered that summarize, serve, and access up-to-date and real-time data.

Generally, these are OLTP systems that provide ACID properties. These systems include, but are not limited to relational databases, NoSQL databases, but also unstructured data.

- Only some of the possible summary views may be materialized.

- **Extract** Data:
  - Get data from multiple, heterogeneous, and external sources.
- **Clean** Data:
  - Detect errors in the data and rectify them if possible.
- **Transform** Data:
  - Convert data from legacy or host format to warehouse format.
- **Load** Data:
  - Sort, summarize, consolidate, compute views, check integrity, and build indexes and partitions.
- **Refresh** Data:
  - Propagate only the updates from the data sources to the warehouse.

Generally speaking:

## Metadata

Data about data.

Three types: *business*, *process execution*, and *technical* metadata. **Business Metadata**

- Business terms and definitions.
- Logical data mapping.
- Data ownership.
- Charging policies.

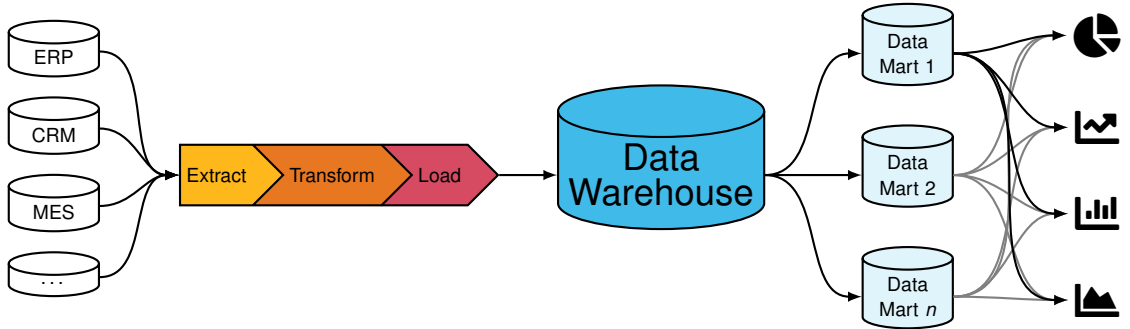
## Process Execution Metadata

- Data acquisition schedule.
- Data-cleaning specifications.
- Aggregate specifications.
- Slowly changing dimensions policies.
- Duration of ETL / rows rejected and successful.

## Technical Metadata

- Table structures and table attributes.
- Derived data definitions.
- Results from data profiling.
- Data lineage.

<sup>3</sup>C.f. chapter 9 of R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*, Wiley, 2004, ISBN: 978-0764567575

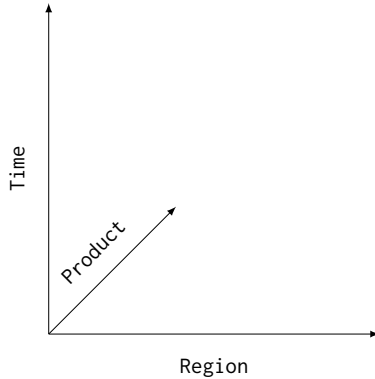


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# Data Warehouse Modeling: Data Cube and OLAP

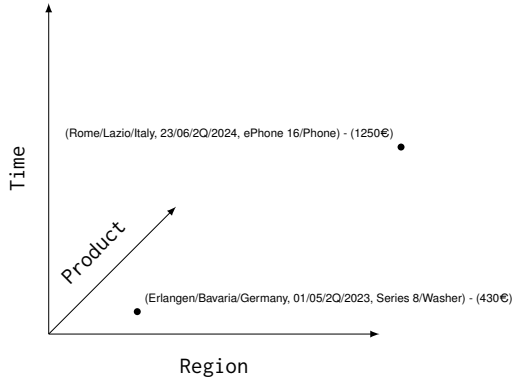


- Data warehouse is based on a **multidimensional data model**
- It views data in the form of a **data cube**.
- A **data cube** contains **two** different kinds of data:
  - **Dimensions:** Information that can be used to group the data.
    - A dimension often comes with different levels of granularity.
    - Example: Time (Granularity levels: day, month, quarter, year).
  - **Facts:** Information that can be aggregated.
    - Example: Price.



## Imagine:

- 3-D coordinate system

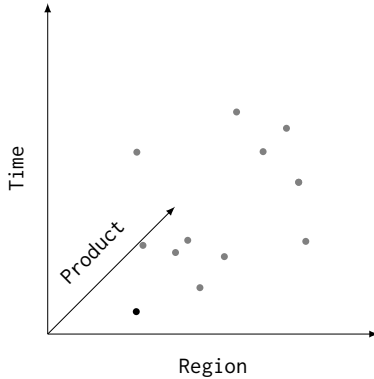


## Dimensional Values:

- Used to locate data points
- Example:  
Erlangen/Bavaria/Germany,  
01/05/2Q/2023,  
Series 8/Washer

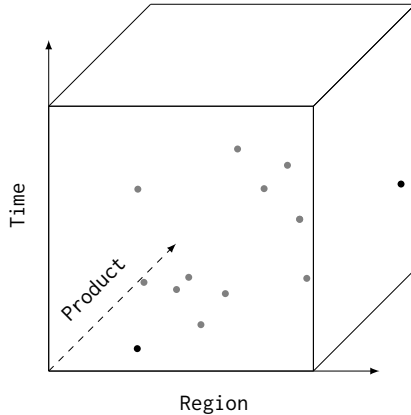
## Facts:

- Used to describe data points
- Example: 430€



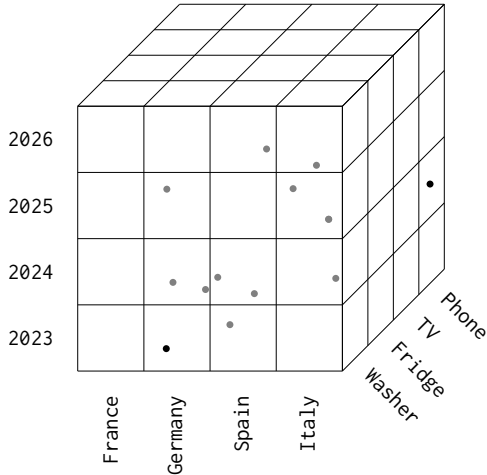
## A Real Data Cube:

- Contains a lot of data points
- Many more than shown ...



## The Data Cube:

- Encapsulates all data points
- Can be used to aggregate all facts

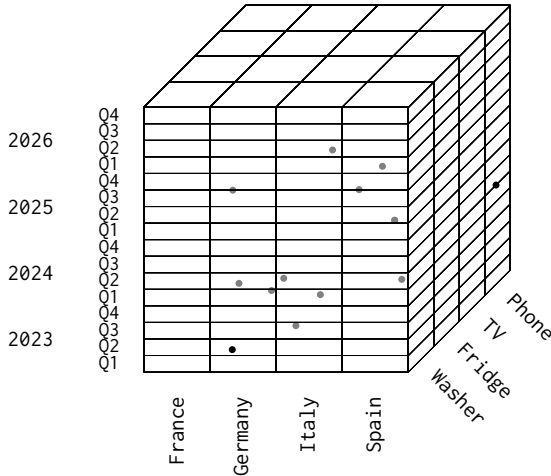


## A Cube of Cubes:

- The cube can be sliced into smaller cubes

## The Smaller Cubes:

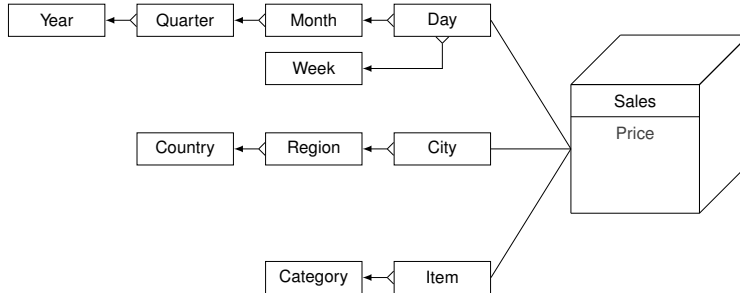
- Can be analyzed separately
- Are in the highest granularity of each dimension



## Even Smaller Cubes:

- Often the cubes can be sliced into even smaller cubes
- By going to the next finer granularity level in at least one dimension

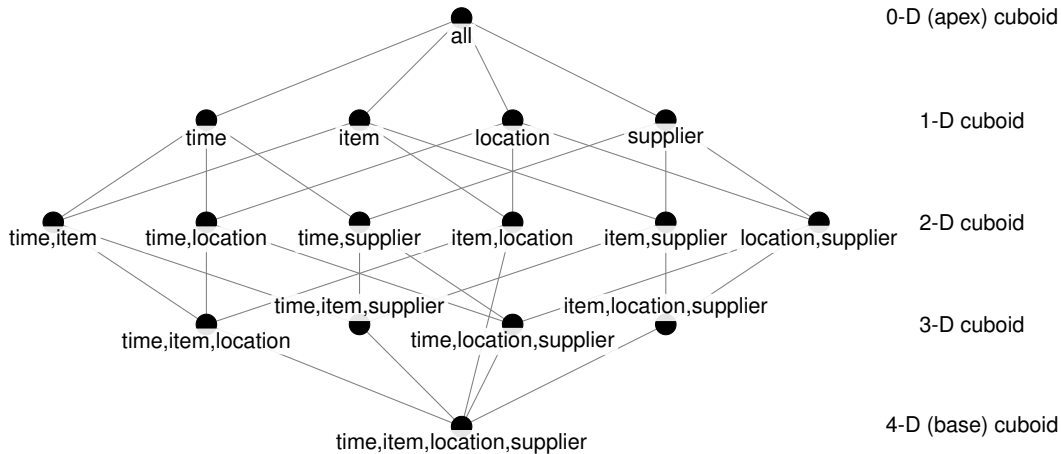
- A **mE/R model**<sup>5</sup> contains both **dimensions** and **facts**.
- Very good in representing **dimensional hierarchies**.



<sup>5</sup>C. Sapia et al., "Extending the E/R model for the multidimensional paradigm," in *Advances in Database Technologies, ER '98 Workshops on Data Warehousing and Data Mining, Mobile Data Access, and Collaborative Work Support and Spatio-Temporal Data Management, Singapore, November 19-20, 1998, Proceedings*, Y. Kambayashi et al., Eds., ser. Lecture Notes in Computer Science, vol. 1552, Springer, 1998, pp. 105–116. doi: 10.1007/978-3-540-49121-7\_9 [Online]. Available: [https://doi.org/10.1007/978-3-540-49121-7\\_9](https://doi.org/10.1007/978-3-540-49121-7_9)



- Each data cube can be aggregated.
- In this process, it is possible to use only individual dimensions for aggregation:
  - **$n$ -dimensional base cube.**
    - Called a base cuboid in data warehousing literature.
  - **Top most 0-dimensional cuboid.**
    - Holds the highest-level of summarization.
    - Called the apex cuboid.
  - **Lattice of cuboids.** (Forms a data cube)



## 1. Star schema:

- A fact table in the middle connected to a set of dimension tables.

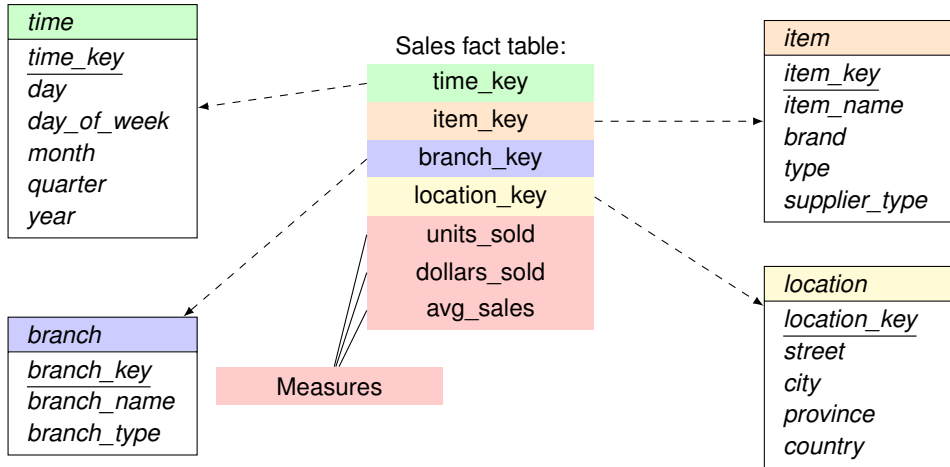
## 2. Snowflake schema:

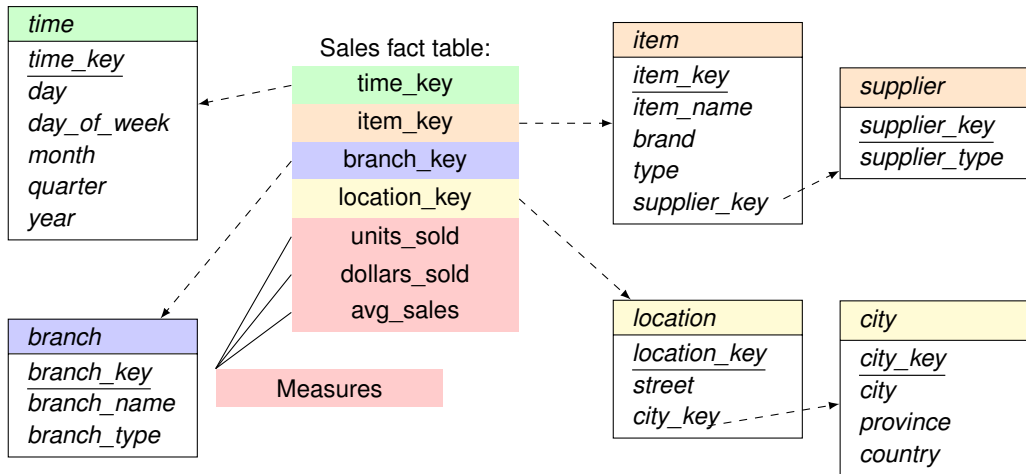
- A refinement of the star schema where some dimensional hierarchies are **normalized** into a set of smaller dimension tables, forming a shape similar to a snowflake.
- I. e. dimension tables of a star schema are split into multiple (dimension) tables along their respective granularity level, but not split/normalized for every granularity.

## 3. Fact constellations:

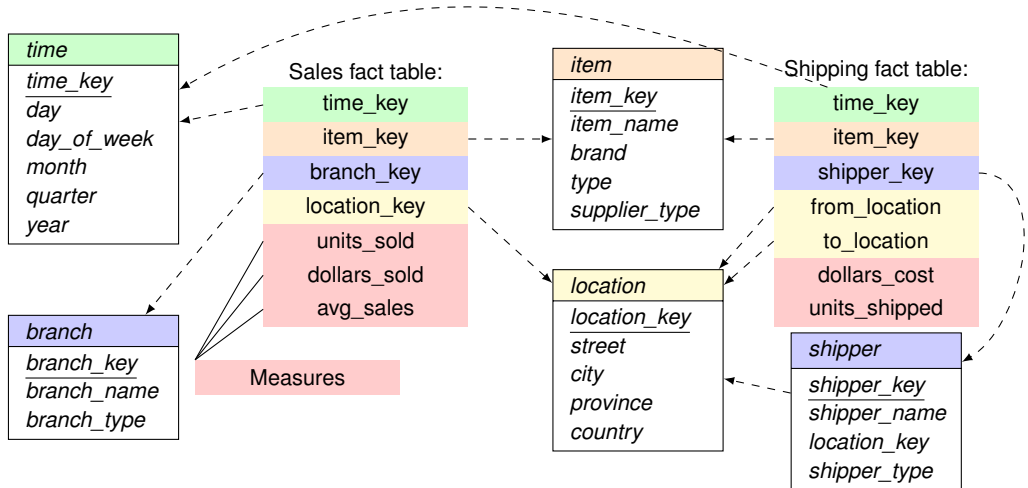
- Multiple fact tables sharing dimension tables, viewed as a collection of stars, therefore called **galaxy schema** or fact constellation.

# Example of a Star Schema





# Example of Fact Constellation



## Data-Cube Measure

A *data-cube measure* is a numeric function that can be evaluated at each point in the data cube space.

### Three Categories:

#### 1. Distributive:

- If the result derived by applying the function to the  $n$  aggregate values obtained for  $n$  partitions of the dataset is the same as that derived by applying the function on all the data without partitioning.  
E.g. COUNT, SUM, MIN, MAX.

#### 2. Algebraic:

- If it can be computed by an algebraic function with  $M$  arguments, each of which is obtained by applying a distributive aggregate function.  
E.g. AVG,  $\text{MIN}_N$ , STD.

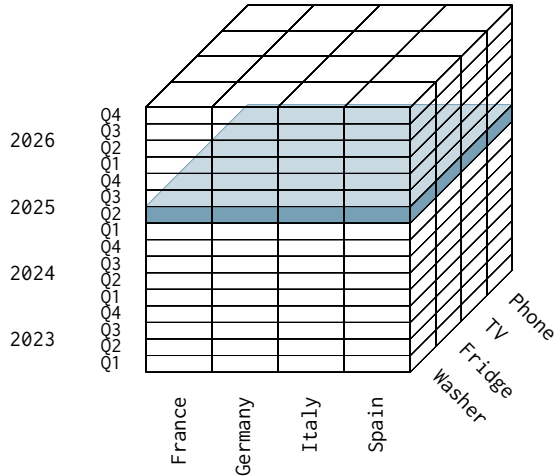
#### 3. Holistic:

- If there is no constant bound on the storage size needed to describe a subaggregate.  
E.g. MEDIAN, MODE, RANK.

- **Non-trivial property.**
  - Next to name and value range.
- **Defines the set of aggregation operations that can be executed on a measure (a fact).**
- **STOCK:** Measure at a specific point in time.
  - Aggregated as desired.
  - E.g. sales turnover, quantity of an item ordered per day.
- **FLOW:** Measure over a period of time.
  - Aggregated as desired, but temporal aggregation not permitted.
  - E.g. total stock and total inventory. Yet, summarization of article stock over multiple days makes no sense!
- **VPU (Value per Unit):** Measures that cannot be summed.
  - E.g. unit price, tax rates, exchange rates.
- **Always applicable: MIN, MAX and AVG.**



- **Slice and dice: project and select.**
  - Selecting only certain dimensions/value ranges from a cube
- **Roll up (drill up): summarize data.**
  - By climbing up hierarchy or by dimension reduction.
- **Drill down (roll down): reverse of roll up.**
  - From higher-level summary to lower-level summary or detailed data, or introducing new dimensions.
- **Pivot (rotate):**
  - Reorient the cube, visualization, 3D to series of 2D planes.
- **Other operations:**
  - **Drill across:** involving (across) more than one fact table.
  - **Drill through:** through the bottom level of the cube to its back-end relational tables (using SQL).



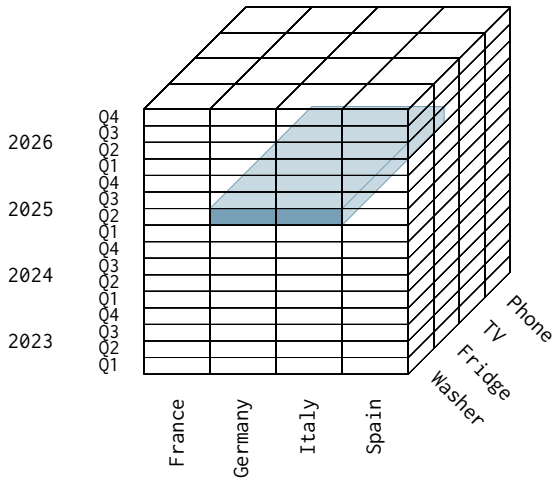
## Slice

### Basic idea:

- Perform a selection on one dimension of the cube.

### Example:

- Select the data for Q2 2025.



## Dice

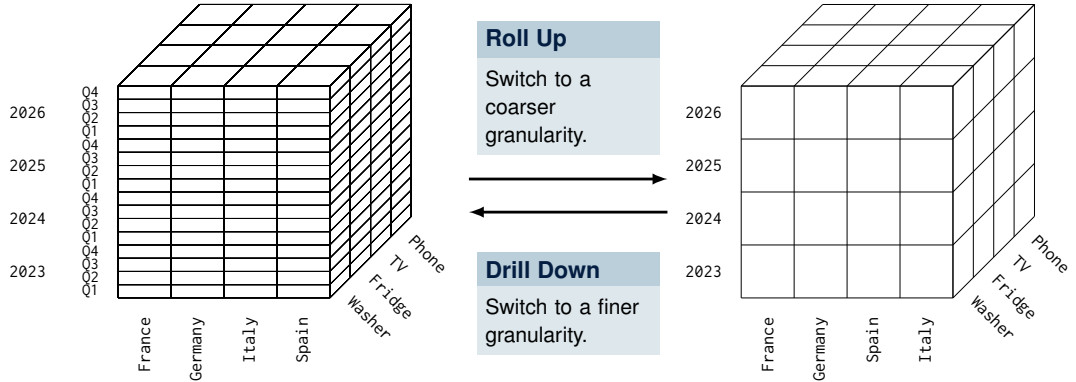
### Basic idea:

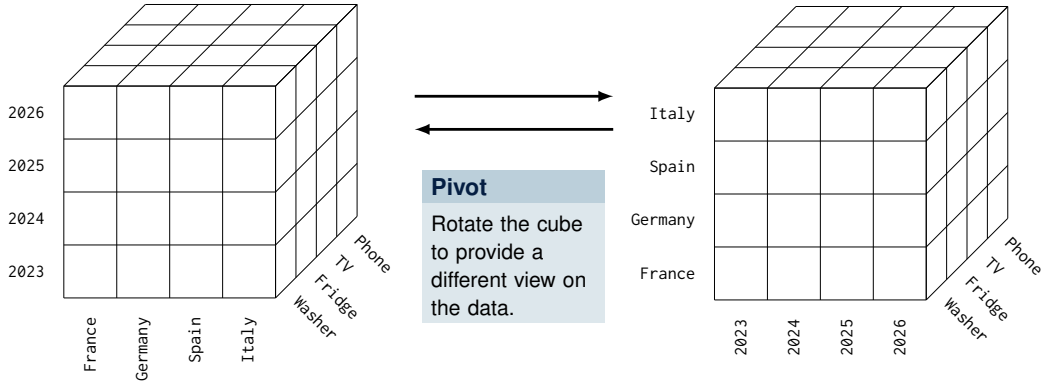
- Perform a selection on more than one dimension.
- It does not have to be on all dimensions!

### Example:

- Select the data for Q2 2025 and the regions Germany and Italy.

# OLAP Operations: Roll Up and Drill Down





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# Data Warehouse Design and Usage

## Four views regarding the design of a data warehouse:

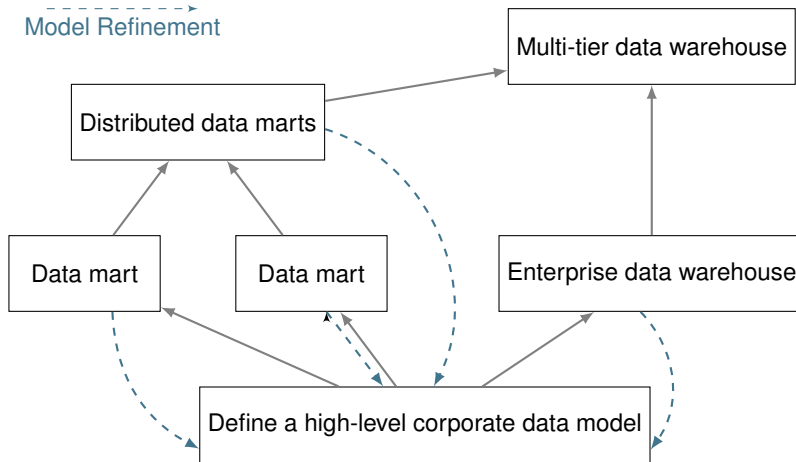
- **Top-down view:**
  - Allows selection of the relevant information necessary for the data warehouse.
- **Data-source view:**
  - Exposes the information being captured, stored, and managed by operational systems.
- **Data warehouse view:**
  - Consists of fact tables and dimension tables.
- **Business-query view:**
  - Sees the perspectives of data in the warehouse from the view of the end-user.

- **Top-down, bottom-up approaches or a combination of both:**
  - **Top-down:** starts with overall design and planning (mature).
  - **Bottom-up:** starts with experiments and prototypes (rapid).
- **From software-engineering point of view:**
  - **Waterfall:** structured and systematic analysis at each step before proceeding to the next.
  - **Spiral:** rapid generation of increasingly functional systems, short turn-around time.
- **Typical Data warehouse design process:**
  1. Choose a **business process** to model, e.g., orders, invoices, etc.
  2. Choose a **grain** (atomic level of data) of the business process.
  3. Choose **dimensions** that will apply to each fact-table record.
  4. Choose a **measure** that will populate each fact-table record.

## DWH Construction is No Easy Feat

Construction of a data warehouse is a difficult long-term task. It is absolutely necessary that its implementation scope is clearly defined at the beginning. Goals and tasks should be *SMART* (specific, measurable, achievable, relevant, and time-related).





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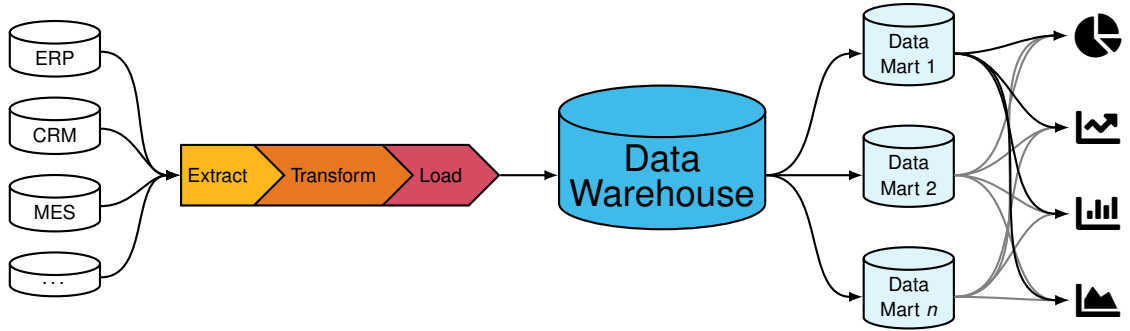
# Data Warehouse and Data Mining

## Why online analytical mining?

- DW contains integrated, consistent, cleaned data.
- Available information-processing structure surrounding data warehouses.
  - ODBC, OLEDB, Web access, service facilities, reporting, and OLAP tools.
- OLAP-based exploratory data analysis.
  - Mining with drilling, dicing, pivoting, etc.
- Online selection of data-mining functions.
  - Integration and swapping of multiple mining functions, algorithms, and tasks.

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<sup>5</sup>OLAM = Online Analytical Mining



- Data mining algorithms in transformation step: E.g. integrate articles from two systems that have different article group hierarchy. Goal: Map one article group hierarchy to the existing article group hierarchy.
- Frequent pattern mining and clustering in reporting: E.g. affinity analysis, revenue prediction, cluster customers and use this insight for a new marketing campaign.

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# Summary

- **Data warehousing: multi-dimensional model of data.**
  - A data cube consists of dimensions and measures.
  - Star schema, snowflake schema, fact constellations.
  - OLAP operations: drilling, rolling, slicing, dicing and pivoting.
- **Data warehouse architecture, design, and usage.**
  - Multi-tiered architecture.
  - Business-analysis design framework.
  - Information processing, analytical processing, data mining, OLAM (Online Analytical Mining).

Our appendix in this document covers:

- **Implementation: efficient computation of data cubes.**
  - Partial vs. full vs. no materialization.
  - Indexing OLAP data: Bitmap index and join index.
  - OLAP query processing.
  - OLAP servers: ROLAP, MOLAP, HOLAP.
- **Data generalization: attribute-oriented induction.**

Additionally, check out these books:

- R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*. Wiley, 2004, ISBN: 978-0764567575
- W. H. Inmon, *Building the Data Warehouse*. Wiley, 2005, 4th edition, ISBN: 978-076459-944-6
- In German: A. Bauer and H. Günzel, *Data Warehouse Systeme – Architektur, Entwicklung, Anwendung*. dpunkt.verlag GmbH, 2004, 4th edition, ISBN: 978-3-89864-785-4

## Any questions about this chapter?

Ask them now or ask them later in our forum:



 [https://www.studon.fau.de/studon/goto.php?target=1code\\_kh3JHJdN](https://www.studon.fau.de/studon/goto.php?target=1code_kh3JHJdN)



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# Appendix

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# Data Warehouse Implementation

- **Data cube can be viewed as a lattice of cuboids.**
- The bottom-most cuboid is the base cuboid.
- The top-most cuboid (apex) contains only one cell.
- How many cuboids in an  $n$ -dimensional cube with  $L_i$  levels associated with dimension  $i$ ?

$$T = \prod_{i=1}^n (L_i + 1). \quad (1)$$

- **Materialization of data cube.**
  - Materialize each (cuboid) (full materialization), none (no materialization), or some (partial materialization).
  - Selection of cuboids to materialize based on size, sharing, access frequency, etc.

- **Cube definition and computation in DMQL:**

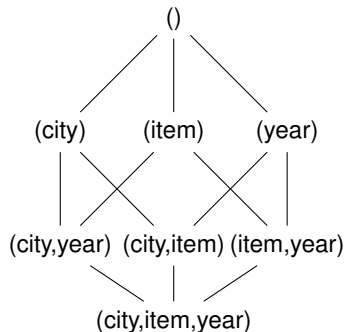
```
DEFINE CUBE sales [item, city, year]:  
SUM (sales_in_dollars);  
COMPUTE CUBE sales;
```

- **Transform it into an SQL-like language:**

```
with a new operator CUBE BY (Gray et al. 96).  
SELECT item, city, year, SUM (amount)  
FROM sales  
CUBE BY item, city, year;
```

- **Need to compute the following GROUP BYs:**

```
(city, item, year),  
(city, item), (city, year),  
(item, year),  
(city), (item), (year)  
( )
```



- Index on a particular column.
- Each value in the column has a bit vector: bit-op is fast.
- Length of bit vector:  $\#$  of records in base table.
- $i$ -th bit set, if  $i$ -th row of base table has value of bit vector.
- Not suitable for high-cardinality domains:
  - A bit compression technique called Word-Aligned Hybrid (WAH) makes it work for high-cardinality domain as well [Wu et al., TODS'06].

**Base table**

| Cust | Region  | Type   |
|------|---------|--------|
| C1   | Asia    | Retail |
| C2   | Europe  | Dealer |
| C3   | Asia    | Dealer |
| C4   | America | Retail |
| C5   | Europe  | Dealer |

**Index on region**

| RecID | Asia | Europe | America |
|-------|------|--------|---------|
| 1     | 1    | 0      | 0       |
| 2     | 0    | 1      | 0       |
| 3     | 1    | 0      | 0       |
| 4     | 0    | 0      | 1       |
| 5     | 0    | 1      | 0       |

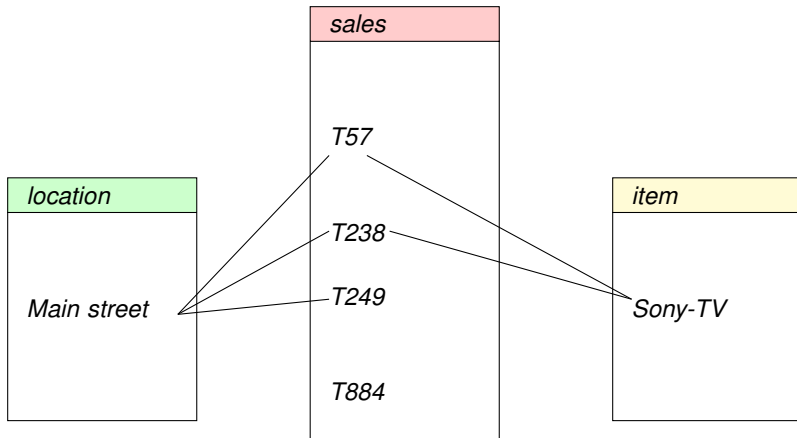
**Index on type**

| RecID | Retail | Dealer |
|-------|--------|--------|
| 1     | 1      | 0      |
| 2     | 0      | 1      |
| 3     | 0      | 1      |
| 4     | 1      | 0      |
| 5     | 0      | 1      |

- **Join index:**

$$JI(R-id, S-id) \quad \text{where} \quad R(R-id, \dots) \bowtie S(S-id, \dots). \quad (2)$$

- **Traditional indices map the values to a list of record ids.**
  - Materializes relational join in JI-file and speeds it up.
- **In data warehouses, join index relates the values of the dimensions of a star schema to rows in the fact table.**
  - E.g. fact table: Sales and two dimensions location and item.
    - A join index on location maintains for each distinct location a list of R-ids of the tuples recording the sales in that location.
  - Join indices can span multiple dimensions.



- **Determine which operations should be performed on the available cuboids.**
  - Transform drill, roll, etc. into corresponding SQL and/or OLAP operations.  
E.g. dice = selection + projection.
- **Determine which materialized cuboid(s) should be selected for OLAP operation.**
  - Let the query to be processed be on {brand, province\_or\_state} with the condition "year = 2004", and there are 4 materialized cuboids available:
    - 1) year, item\_name, city.
    - 2) year, brand, country.
    - 3) year, brand, province\_or\_state.
    - 4) item\_name, province\_or\_state where year = 2004.
  - Which should be selected to process the query?
- **Explore indexing structures and compressed vs. dense-array structures in MOLAP.**



- **Relational OLAP (ROLAP).**

- Use relational or extended-relational DBMS to store and manage warehouse data and OLAP middleware.
- Include optimization of DBMS backend, implementation of aggregation navigation logic, and additional tools and services.
- Greater scalability.

- **Multidimensional OLAP (MOLAP).**

- Sparse array-based multidimensional storage engine.
- Fast indexing to pre-computed summarized data.

- **Hybrid OLAP (HOLAP) (e.g., Microsoft SQL-Server).**

- Flexibility, e.g., low level: relational, high-level: array.

- **Specialized SQL servers (e.g., Redbricks).**

- Specialized support for SQL queries over star/snowflake schemas.

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# Attribute-Oriented Induction

- **Summarize data:**
  - **By replacing relatively low-level values with higher-level concepts**  
e.g. numerical values for the attribute age  
e.g. young, middle-aged and senior.
  - **By reducing the number of dimensions**  
e.g. removing birth\_date and telephone\_number  
when summarizing the behavior of a group of students.
  - Describe concepts in concise and succinct terms at generalized (rather than low) levels of abstractions:
    - Facilitates users in examining the general behavior of the data.
    - Makes dimensions of a data cube easier to grasp.

- **Proposed in 1989** (KDD'89 workshop).
- **Not confined to categorical data nor to particular measures.**
- **How is it done?**
  - Collect the **task-relevant data** (initial relation) using a relational database query.
  - Perform **generalization** by attribute removal or attribute generalization.
  - Apply **aggregation** by merging identical, generalized tuples and accumulating their respective counts.
  - Interaction with users for knowledge presentation.

- **Example:** Describe general characteristics of graduate students in a university database.
- **Step 1:** Fetch relevant set of data using an SQL statement, e.g.  

```
SELECT name, gender, major, birth_place, birth_date, residence, phone#, gpa  
FROM student  
WHERE student_status IN "Msc", "MBA", "PhD";
```
- **Step 2:** Perform attribute-oriented induction.
- **Step 3:** Present results in generalized-relation, cross-tab, or rule forms.

| Name           | Gender   | Major         | Birth place           | Birth date | Residence                | Phone number | GPA          |
|----------------|----------|---------------|-----------------------|------------|--------------------------|--------------|--------------|
| Jim            | M        | CS            | Vancouver, BC, Canada | 08-21-76   | 3511 Main St., Richmond  | 687-4598     | 3.67         |
| Scott Lachance | M        | CS            | Montreal, Que, Canada | 28-07-75   | 345 1st Ave., Richmond   | 253-9106     | 3.70         |
| Laura Lee      | F        | Physics       | Seattle, WA, USA      | 25-08-70   | 125 Austin Ave., Burnaby | 420-5232     | 3.83         |
| Removed        | Retained | Sci, Eng, Bus | Canada, Foreign       | Age range  | City                     | Removed      | Excl, Vg,... |

| Gender | Major   | Birth re-<br>gion | Age range | Residence | GPA       | Count |
|--------|---------|-------------------|-----------|-----------|-----------|-------|
| M      | Science | Canada            | 20-25     | Richmond  | Very good | 16    |
| F      | Science | Foreign           | 25-30     | Burnaby   | Excellent | 22    |
| ...    | ...     | ...               | ...       | ...       | ...       | ...   |

Cross-table of birth region and gender:

|       | Canada | Foreign | Total |
|-------|--------|---------|-------|
| M     | 16     | 14      | 30    |
| F     | 10     | 22      | 32    |
| Total | 26     | 36      | 62    |



- **Data focusing:**
  - Task-relevant data, including dimensions.
  - The result is the **initial relation**.
- **Attribute removal:**
  - Remove attribute A, if there is a large set of distinct values for A, but (1) there is no generalization operator on A, or (2) A's higher-level concepts are expressed in terms of other attributes.
- **Attribute generalization:**
  - If there is a large set of distinct values for A, and there exists a **set of generalization operators** on A, then select an operator and generalize A.
- **Attribute-threshold control:**
  - Typical 2-8, specified/default.
- **Generalized-relation-threshold control:**
  - Control the final relation/rule size.

- **InitialRel:**
  - Query processing of task-relevant data, deriving the initial relation.
- **PreGen:**
  - Based on the analysis of the number of distinct values in each attribute, determine generalization plan for each attribute: removal? Or how high to generalize?
- **PrimeGen:**
  - Based on the PreGen plan, perform generalization to the right level to derive a "prime generalized relation", accumulating the counts.
- **Presentation:**
  - User interaction:
    1. Adjust levels by drilling.
    2. Pivoting.
    3. Mapping into rules, cross tabs, visualization presentations.

- **Generalized relation:**

- Relations where some or all attributes are generalized, with counts or other aggregation values accumulated.

- **Cross tabulation:**

- Mapping results into cross-tabulation form (similar to contingency tables).
- Visualization techniques: pie charts, bar charts, curves, cubes, and other visual forms.

- **Quantitative characteristic rules:**

- Mapping generalized results into characteristic rules with quantitative information associated with it, e.g.

$$\text{grad}(x) \wedge \text{male}(x) \implies \quad (3)$$

$$\text{birth\_region}(x) = \text{"Canada"}[t : 53\%] \vee \quad (4)$$

$$\text{birth\_region}(x) = \text{"foreign"}[t : 47\%]. \quad (5)$$

- **Comparison: Comparing two or more classes.**
- **Method:**
  - Partition the set of relevant data into the **target class** and the **contrasting class(es)**.
  - Generalize both classes to the same high-level concepts (i.e. AOI).
    - Including aggregation.
  - Compare tuples with the same high-level concepts.
  - Present for each tuple its description and two measures.
    - Support – distribution within single class (counts, percentage).
    - Comparison – distribution between classes.
  - Highlight the tuples with strong discriminant features.
- **Relevance Analysis:**
  - Find attributes (features) which best distinguish different classes.

- **Similarity:**

- Data generalization.
- Presentation of data summarization at multiple levels of abstraction.
- Interactive drilling, pivoting, slicing and dicing.

- **Differences:**

- OLAP has systematic preprocessing, query independent, and can drill down to rather low level.
- AOI has automated desired-level allocation and may perform dimension-relevance analysis/ranking when there are many relevant dimensions.
- AOI works on data which are not in relational forms.